

Week 11: Indexed terms and typed evaluation

Functional Programming in Lean

Learning goals

- ▶ Read a type indexed by a value
- ▶ Construct expressions that preserve their result type
- ▶ Explain why an ill-typed expression has no constructor

Central idea

The index of Expr records the type of the value produced by evaluation.

Lean example

```
inductive Expr : Ty → Type where
  | number (n : Nat) : Expr .nat
  | add (x y : Expr .nat) : Expr .nat
  | choose (c : Expr .bool)
    (x y : Expr t) : Expr t
```

What to notice

- ▶ `denote` interprets a syntax type as a Lean type
- ▶ Pattern matching refines the result type
- ▶ An `add` node cannot take a `Boolean` child

Live coding

1. Complete the indexed evaluator
2. Construct a conditional expression
3. Try to construct an invalid addition and read the error

Pause and predict

Why does `eval` need a result type depending on `t`?

Exercise connection

Open `exercises/Week11.md` after the lecture.

The complete example is `Course/Week11.lean`; the fill-in version is `Templates/Week11.lean`.